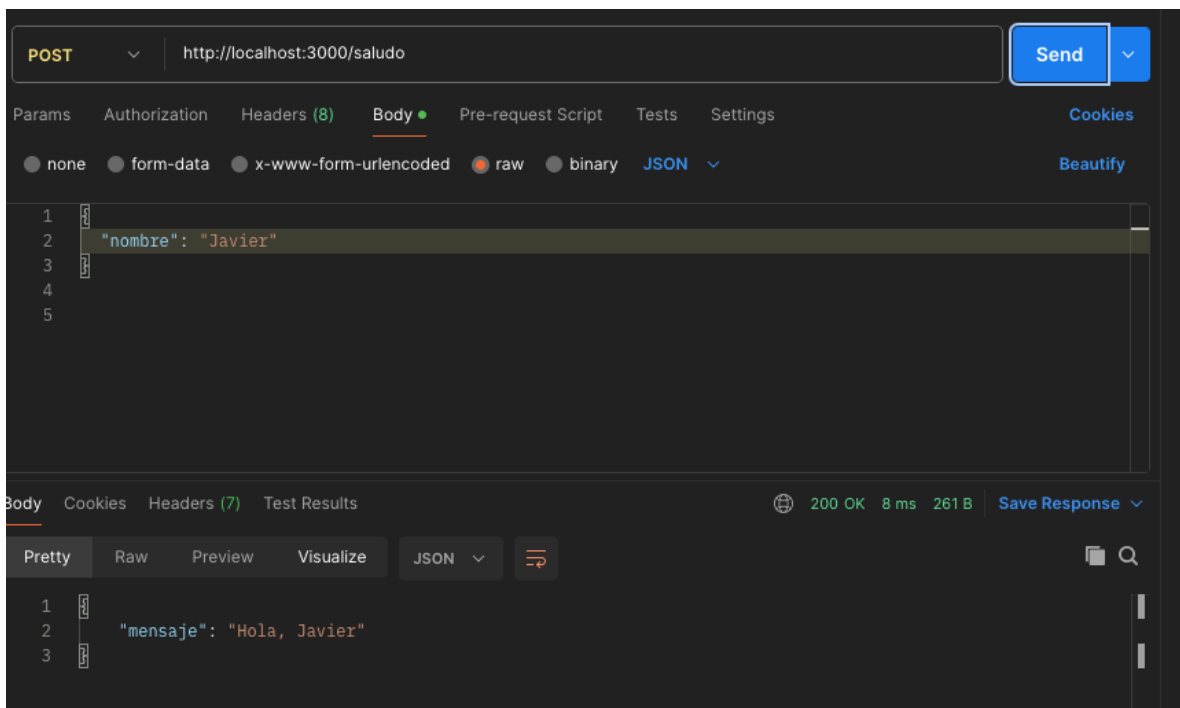


Aplicaciones distribuidas

Evidencia practica 4.

a. Servicio de saludo básico

```
app.post("/saludo", (req, res) => {  
  const { nombre } = req.body;  
  res.json({  
    mensaje: `Hola, ${nombre}`  
  });  
});
```



b. Calculadora de operaciones básicas

```
app.post("/calcular", (req, res) => {  
  const { a, b, operacion } = req.body;  
  let resultado;  
  
  switch (operacion) {  
    case "suma":
```

```

        resultado = a + b;
        break;
    case "resta":
        resultado = a - b;
        break;
    case "multiplicacion":
        resultado = a * b;
        break;
    case "division":
        if (b === 0) {
            return res.status(400).json({
                error: "No se puede dividir entre cero"
            });
        }
        resultado = a / b;
        break;
    default:
        return res.status(400).json({
            error: "Operación inválida"
        });
    }

    res.json({ resultado });
});

```

The screenshot shows a REST client interface with the following details:

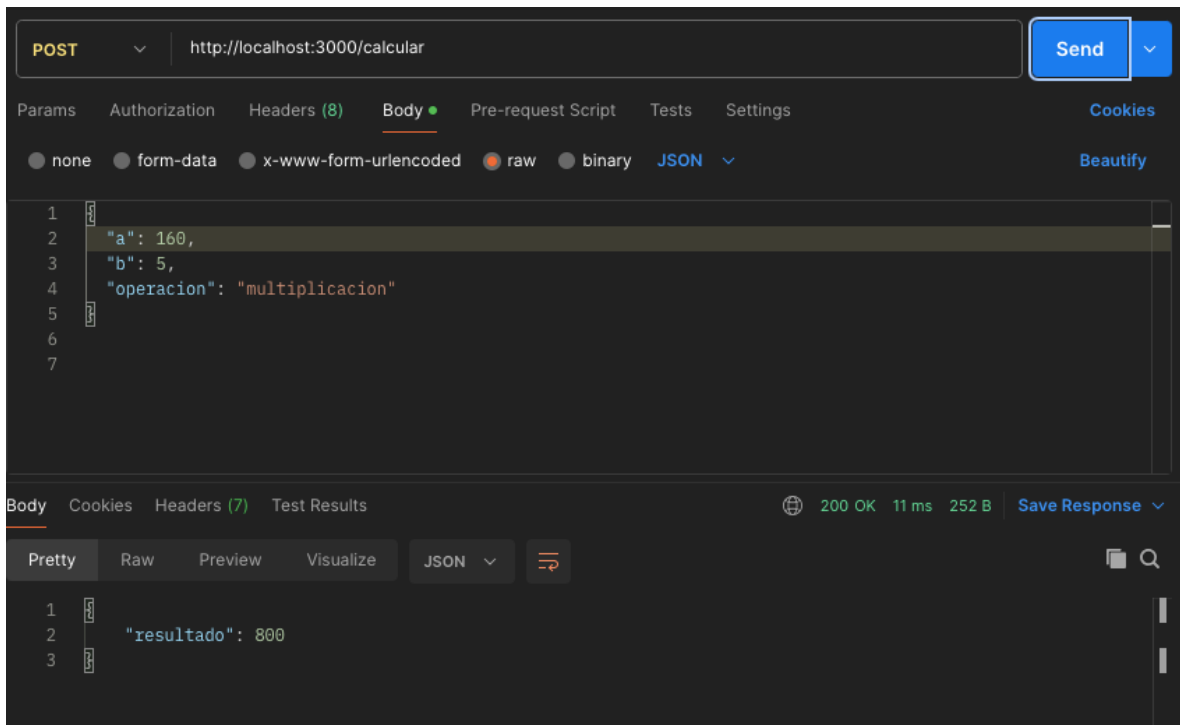
- Method:** POST
- URL:** http://localhost:3000/calculador
- Body Tab:** Selected, showing a JSON object:


```

      {
        "a": 10,
        "b": 0,
        "operacion": "division"
      }
      
```
- Response Tab:** Selected, showing a 400 Bad Request status with the error message:


```

      {
        "error": "No se puede dividir entre cero"
      }
      
```
- Response Details:** 400 Bad Request, 89 ms, 286 B.



c. Gestor de tareas

```
app.post("/tareas", (req, res) => {
  const tareas = leerTareas();
  tareas.push(req.body);
  guardarTareas(tareas);
  res.json({ mensaje: "Tarea creada" });
});

app.get("/tareas", (req, res) => {
  const tareas = leerTareas();
  res.json(tareas);
});

app.put("/tareas/:id", (req, res) => {
  const id = parseInt(req.params.id);
  const tareas = leerTareas();

  const index = tareas.findIndex(t => t.id === id);
  tareas[index] = { ...tareas[index], ...req.body };

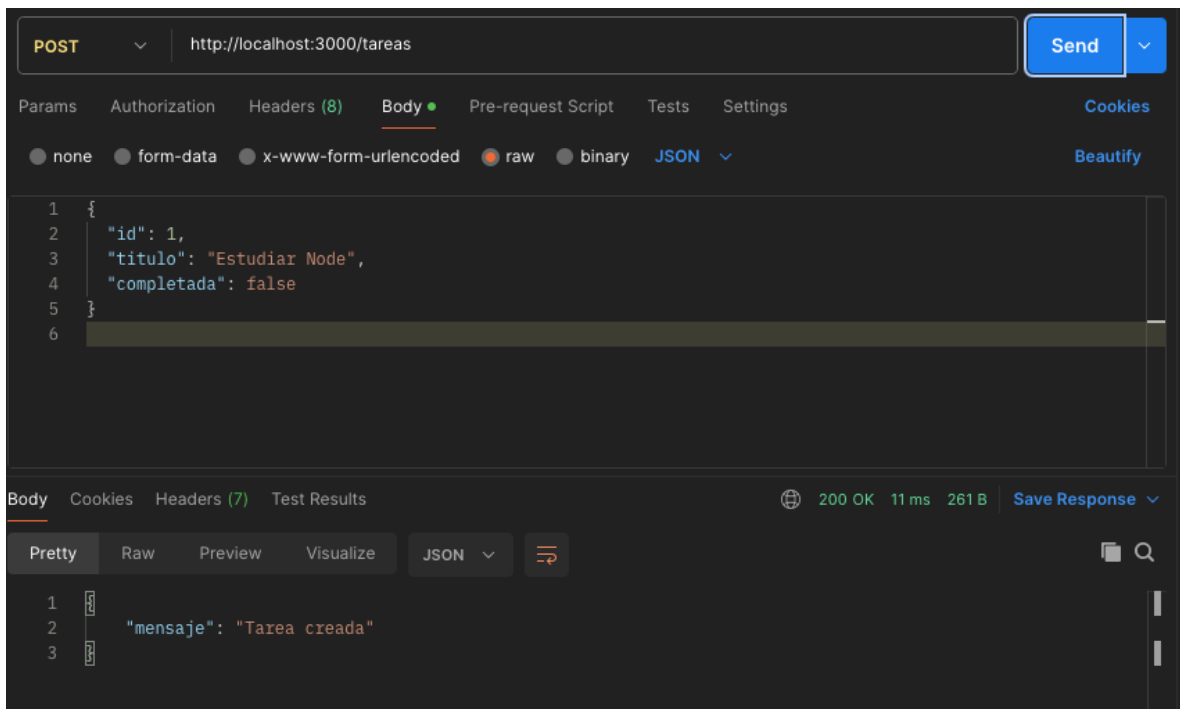
  guardarTareas(tareas);
  res.json({ mensaje: "Tarea actualizada" });
});
```

```
});

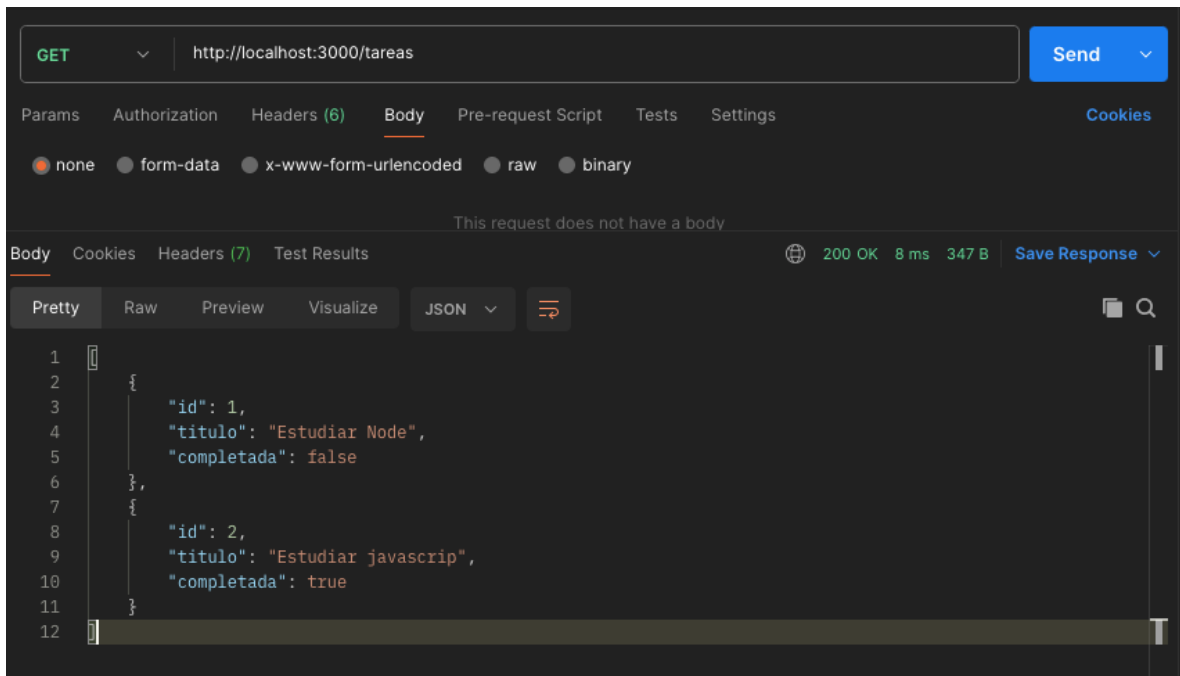
app.delete("/tareass/:id", (req, res) => {
  const id = parseInt(req.params.id);
  const tareas = leerTareas();
  const nuevas = tareas.filter(t => t.id !== id);

  guardarTareas(nuevas);
  res.json({ mensaje: "Tarea eliminada" });
});
```

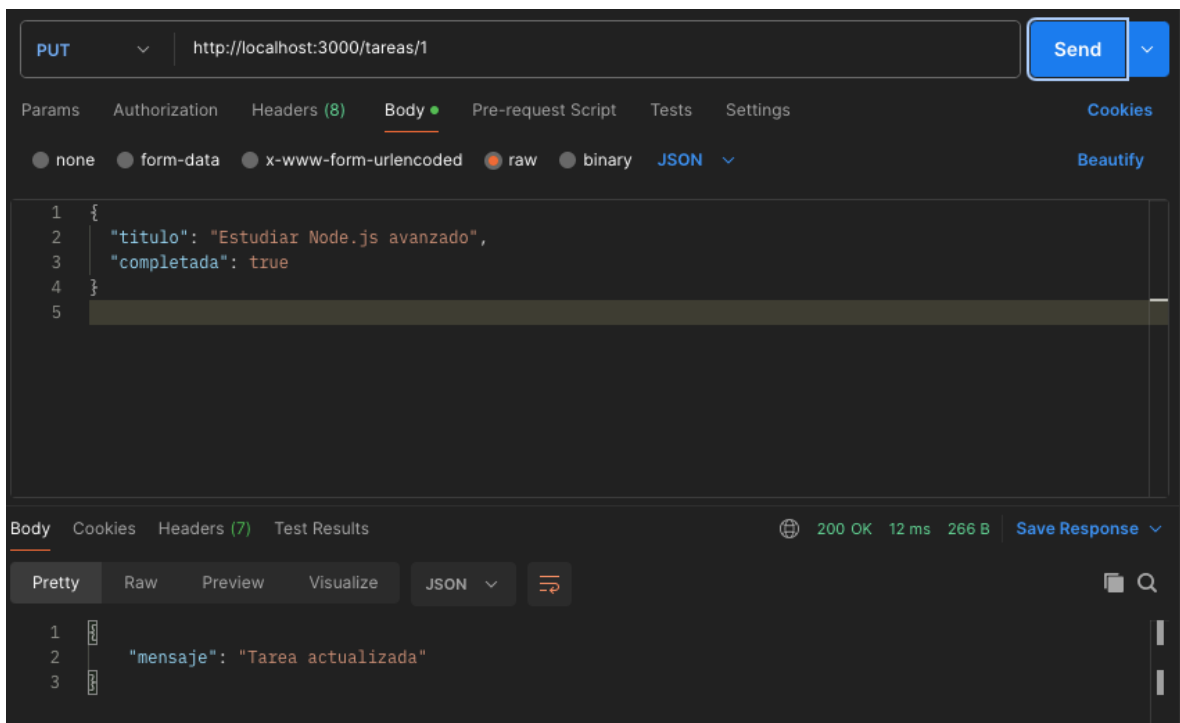
a. Crear tarea



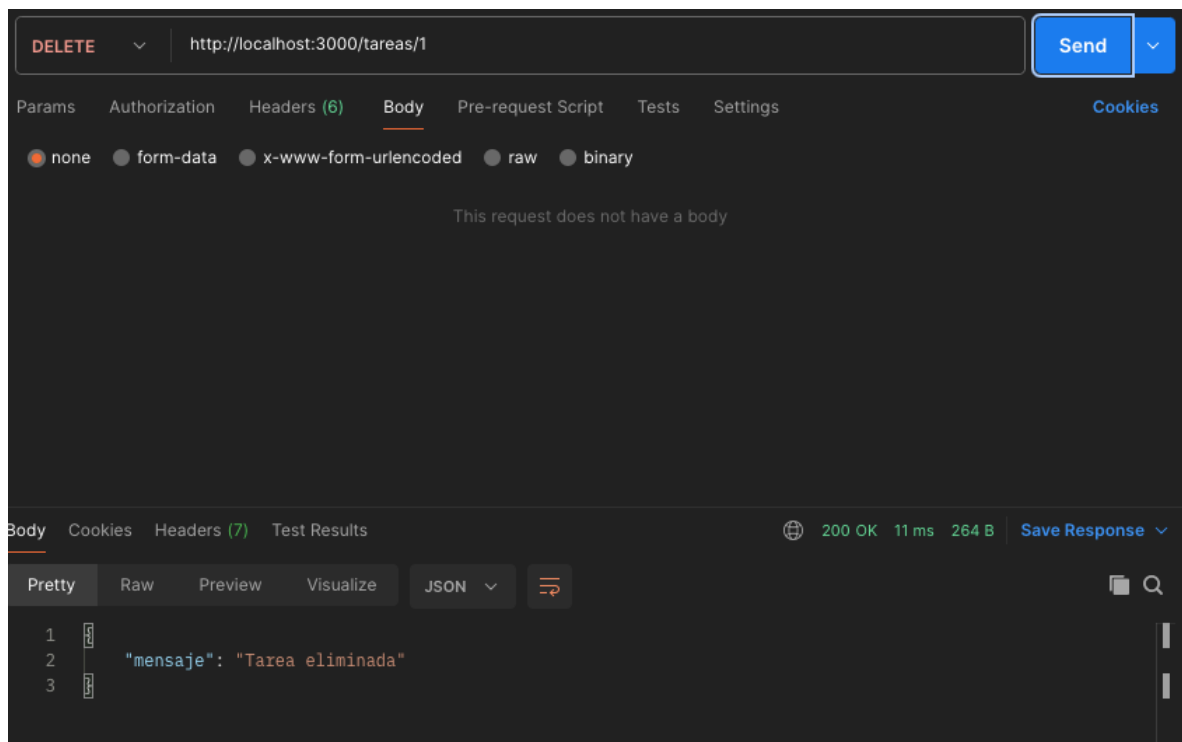
b. Listar tareas



c. Actualizar una tarea

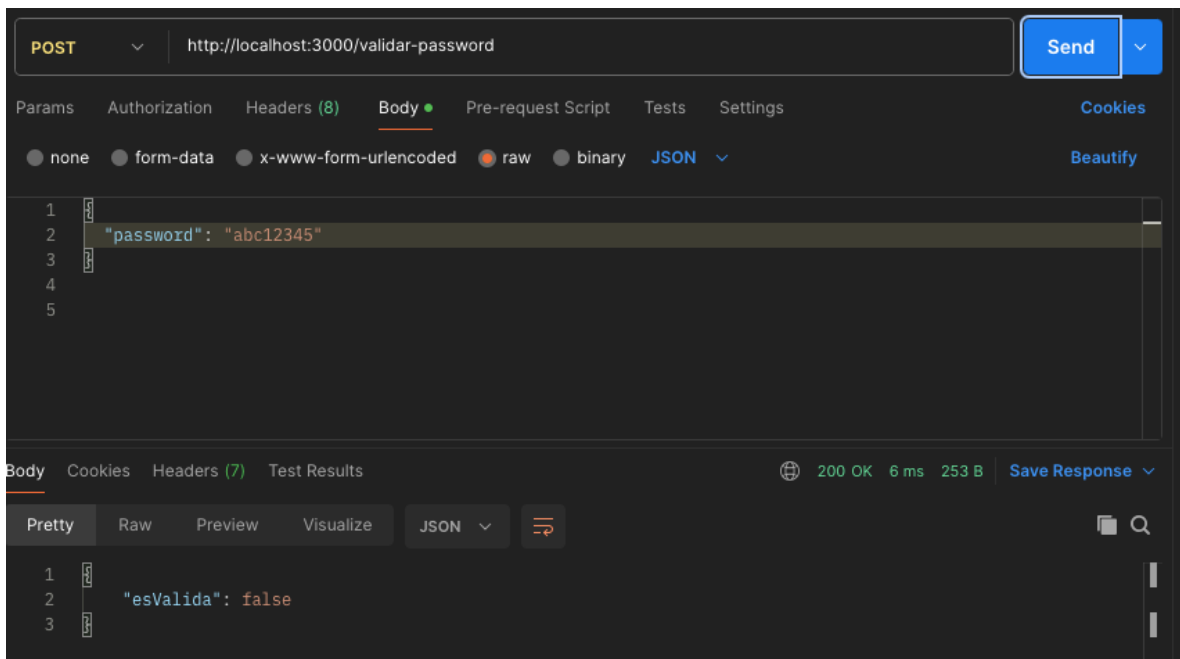
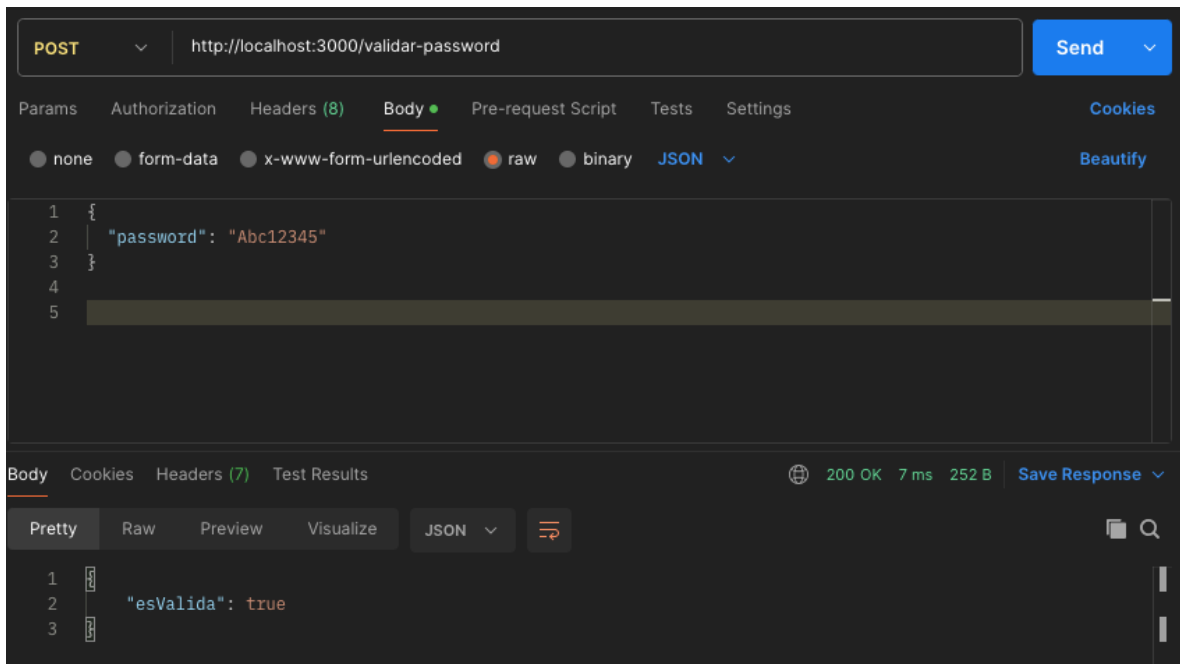


d. Eliminar tarea



d. Validar password

```
app.post("/validar-password", (req, res) => {  
  const { password } = req.body;  
  
  const esValida =  
    password.length >= 8 &&  
    /[A-Z]/.test(password) &&  
    /[a-z]/.test(password) &&  
    /[0-9]/.test(password);  
  
  res.json({ esValida });  
});
```



e. Conversor de temperatura

```
app.post("/convertir-temperatura", (req, res) => {
  let { valor, desde, hacia } = req.body;
  let celsius;

  if (desde === "C") celsius = valor;
  if (desde === "F") celsius = (valor - 32) * 5 / 9;
```

```

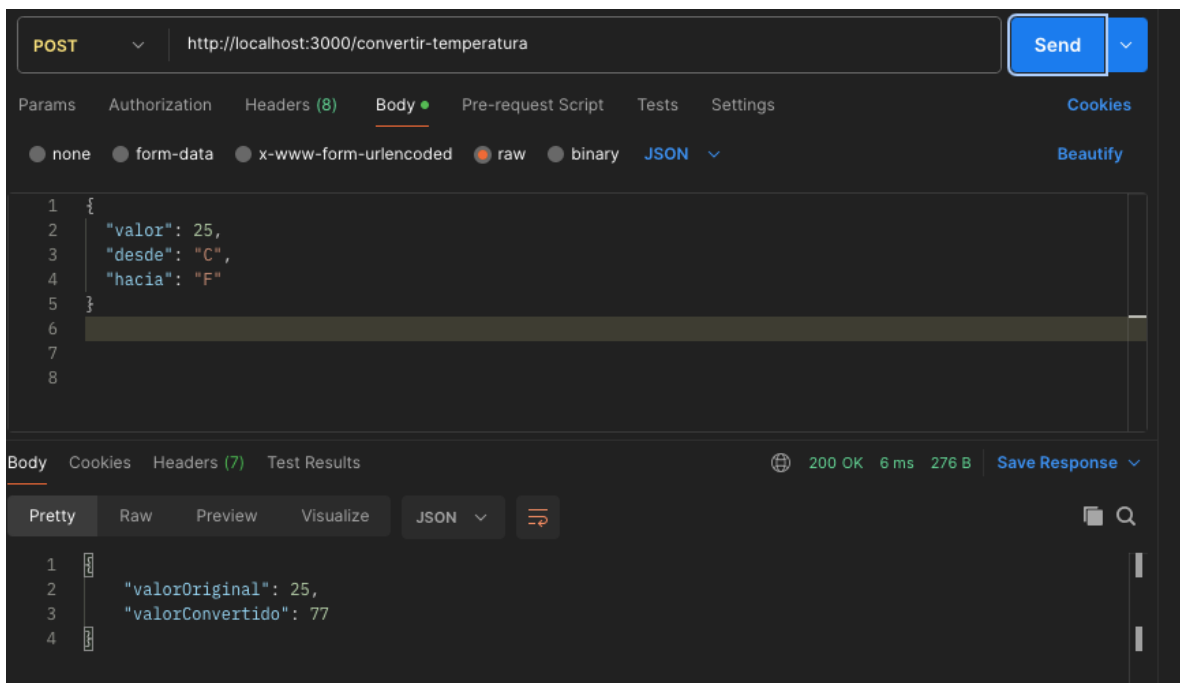
    if (desde === "K") celsius = valor - 273.15;

    let convertido;

    if (hacia === "C") convertido = celsius;
    if (hacia === "F") convertido = celsius * 9 / 5 + 32;
    if (hacia === "K") convertido = celsius + 273.15;

    res.json({
      valorOriginal: valor,
      valorConvertido: convertido
    });
  });
});

```



f. Buscador en Array

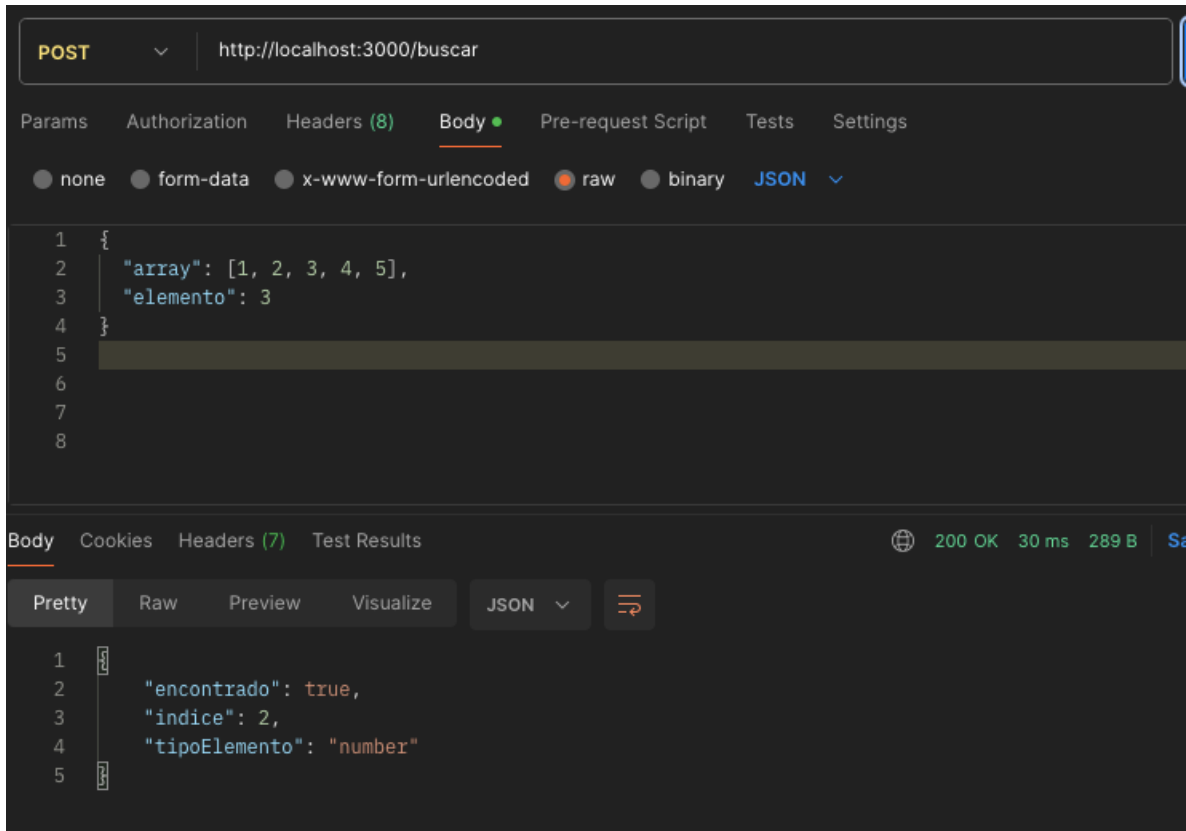
```

app.post("/buscar", (req, res) => {
  const { array, elemento } = req.body;
  const indice = array.indexOf(elemento);

  res.json({
    encontrado: indice !== -1,
    indice: indice,
    tipoElemento: typeof elemento
  });
});

```

```
});
```



g. Contador de palabras

```
app.post("/contar-palabras", (req, res) => {  
  const { texto } = req.body;  
  
  const palabras = texto.trim().split(/\s+/);  
  const unicas = new Set(palabras);  
  
  res.json({  
    totalPalabras: palabras.length,  
    totalCaracteres: texto.length,  
    palabrasUnicas: unicas.size  
  });  
});
```

POST

http://localhost:3000/contar-palabras

Params

Authorization

Headers (8)

Body

Pre-request Script

Tests

Settings

none

form-data

x-www-form-urlencoded

raw

binary

JSON

```
1 {
2   "texto": "Hola mundo en aplicaciones distribudas"
3 }
4
5
6
7
```

Body

Cookies

Headers (7)

Test Results

200 OK

8 ms

294 B

S

Pretty

Raw

Preview

Visualize

JSON

```
1 {
2   "totalPalabras": 5,
3   "totalCaracteres": 38,
4   "palabrasUnicas": 5
5 }
```